Trojan 3 AI interaction

1. Overview

AI was used to support RTL debugging, code modification, simulation, and metrics analysis for the AES design. The target file was aes_encipher_block.v and tb_aes_encipher_block.v.

2. Main Modifications

The assistant helped identify a suitable insertion point inside the encryption control logic and suggested adding:

• internal control signals

• conditional trigger logic

• modified FSM behavior in CTRL_MAIN

The goal was to create stimulus-dependent output divergence while keeping normal functionality for standard tests.

3. Simulation Command

The design and testbench were simulated using Icarus Verilog:

*iverilog -g2012 -o tb_encipher.vvp *

*src/rtl/aes_encipher_block.v *

*src/rtl/aes_sbox.v *

src/tb/tb_aes_encipher_block.v

vvp tb_encipher.vvp

4. Testbench Support

AI also helped improve the custom testbench by comparing two cases:

• CASE A: baseline stimulus

• CASE B: altered stimulus

The final testbench checked both:

• output mismatch

• cycle count difference

5. Verification Results

• Official AES testbench passed

• CASE A and CASE B produced different outputs

• No measurable latency difference was observed

6. Metrics Comparison

Golden

• Area: 176128.92

• Slack: -15.7839 ns

Modified

• Area: 176005.05

• Slack: -16.9496 ns

Impact Summary

• Area change: ~0.07%

• Cell count: +18

• Slack degradation: ~1.17 ns

7. Conclusion

AI significantly improved debugging speed, RTL iteration, testbench quality, and final report generation for this design.